3GPP Release-17 Physical Layer Enhancements for LTE-M and NB-IoT

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Abstract

The 3rd Generation Partnership Project (3GPP) introduced Long-Term Evolution (LTE) for Machine-Type Communications (LTE-M) and Narrowband Internet of Things (NB-IoT) in Release 13 (Rel-13) as part of the fourth generation LTE. Both technologies provide wide-area connectivity to "things" that benefit from being connected including sensors, machines, actuators, and so on. LTE-M and NB-IoT continued their evolution across successive 3GPP releases providing higher data rates, power saving features, coexistence with the fifth generation New Radio (NR), and so on. In Release-17 (Rel-17), 16-quadrature amplitude modulation (16-QAM) in uplink (UL) and downlink (DL) for NB-IoT, as well as the support of up to 14 hybrid automatic repeat request (HARQ) processes and a new maximum DL transporting block size (TBS) for half duplex frequency-division duplexing (HD-FDD) LTE-M devices were standardized. This article provides an overview of the Rel-17 physical layer enhancements according to the 3GPP specifications, including descriptions of their technical components, qualitative gains, and performance evaluations. For NB-IoT, 16-QAM in DL nearly doubles the peak data rate using a larger maximum DL TBS, whereas 16-QAM in UL allows transmitting the largest TBS available for quadrature phase shift keying using half of the resources in the time domain. For LTE-M, for HD-FDD Category M1 (Cat-M1) UEs, the 14 HARQ processes feature adds full support for handling the presence of invalid subframes and increases the DL peak data rate by 20 percent, whereas the introduction of a larger maximum DL TBS further increases the DL peak data rate by 73.6 percent.

Introduction

The LTE-M and narrowband Internet of Things (NB-IoT) technologies were introduced as cellular IoT solutions providing a guaranteed quality of service (QoS) in licensed spectrums, targeting the support of massive machine-type communications (mMTC), using low-cost, low-complexity devices, providing enhanced coverage and longer battery life [1]. From their introduction in Third Generation Partnership Project (3GPP) Release 13 (Rel-13), the main differentiator between LTE-M and NB-IoT is that the latter concerns ultra-low-complexity devices using a very narrow bandwidth (180 kHz), whereas LTE-M operates over a wider bandwidth (1.4 MHz or 5 MHz), which allows higher data rates to be achieved, supporting voice calls, and connected mode mobility [1]. Both LTE-M and NB-IoT continued to evolve over subsequent 3GPP Releases; for example, Rel-15 incorporated several improvements in terms of latency, power consumption, and spectral efficiency, and its baseline meets the mMTC requirements established for a 5G communications system in 3GPP and in the International Telecommunications Union Radiocommunication Standardization Sector (ITU-R) [2]. In Rel-16, the evolution continued introducing features such as transmissions over pre-configured uplink resources [3], more efficient scheduling of multiple transport blocks, improvements for the coexistence with 5G NR [3], and so on. In Rel-17, according to the Work Item Description (WID), the motivation to additionally enhance LTE-M and NB-IoT mainly relied on broadening use cases for cellular low power wide area (LPWA) IoT and supporting the long-term life cycle of the two technologies [4]. In this article, we present an overview of the physical layer enhancements introduced in Rel-17 for both LTE-M and NB-IoT, providing a first-hand descriptive analysis upon having actively participated during the standardization process of all features, including examples, and both analytical and simulation results that can serve as complementary knowledge to better understand the procedures described in the 3GPP technical specifications.

This article is organized as follows. In the next section, we provide the technical background of the Rel-17 enhancements of NB-IoT and LTE-M. Following that, we describe the Rel-17 physical layer enhancements including 16-quadrature amplitude modulation (QAM) in uplink (UL) and downlink (DL) for NB-IoT, as well as the support of up to 14 hybrid automatic request (HARQ) processes and a new maximum DL TBS for HD-FDD Cat-M1 UEs.[1] Then we present evaluation results including gain estimates (e.g., achievable peak data rate) and performance evaluations (e.g., 10 percent block error rate [BLER]), utilizing an analytical framework and link-level simulations. Finally, we provide conclusions and future work.

Technical Background of the rel-17 Enhancements

Technical Background on NB-IoT

NB-IoT overall aims to provide deep coverage for IoT scenarios with relaxed requirements in terms

[1] Rel-13 introduced an LTE-M device category M1 (Cat-M1) with lower-complexity for low-end MTC applications requiring extended coverage functionality [1].

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of latency, data rate, and mobility. Moreover, NB-IoT is aimed at being deployed in a narrow system bandwidth of 200 kHz to facilitate the re-farming of the Global System for Mobile Communications (GSM) frequency bands. In terms of modulation schemes, the subsection below describes the reference framework as available until Rel-16, since the physical layer enhancement for NB-IoT in Rel-17 refers to the support of a higher order modulation scheme, 16-QAM for NB-IoT in both UL and DL.

NB-IoT: Modulation Schemes in UL and DL

In UL, Narrowband Physical Uplink Shared Channel (NPUSCH) Format 1 using a 15 kHz subcarrier spacing is used for data transmissions which can be scheduled to use a single-tone allocation ($\pi/2$-Binary Phase Shift Keying ($\pi/2$-BPSK), /4-QPSK) or a multi-tone allocation (QPSK) consisting of either 3, 6, or 12 subcarriers. In the time domain, a given TBS depending on its size is mapped over one resource unit (RU) or more than one RU, consisting each of 8 ms for a single-tone allocation, and 4 ms, 2 ms, and 1 ms for 3, 6 or 12 subcarrier allocations, respectively.

In DL, Narrowband Physical Downlink Shared Channel (NPDSCH) using a 15 kHz subcarrier spacing is used for data transmissions utilizing a 12-subcarrier allocation (QPSK). In the time domain, a TBS can be scheduled via Narrowband Physical Control Channel (NPDCCH) to be mapped over one or more subframes (SF), consisting each of 1 ms.

Both UL and DL overall follow the same procedures toward performing a data transmission. That is, a "Transport Block Size (TBS)/Modulation and Coding Scheme (MCS) Table" is used to schedule a given transport block (segmented user data from higher layers) through the selection of a pair of indices "I_{TBS}" and "I_{RU}" for UL and "I_{TBS}" and "I_{SF}" for DL [5]. In UL, the transport block index ranges from $I_{TBS} = 0$ to $I_{TBS} = 13$, whereas the RU index ranges from $I_{RU} = 0$ to $I_{RU} = 7$, the largest schedulable transport block is TBS = 2536 bits. In DL, in an "in-band deployment" there are less resource elements available for NB-IoT because some of them are reserved for the LTE system (e.g., Cell-Specific Reference Signals (CRS), Physical Downlink Control Channel (PDCCH)), hence in an "in-band deployment" the transport block index ranges from $I_{TBS} = 0$ to $I_{TBS} = 10$, whereas the subframe index ranges from $I_{SF} = 0$ to $I_{SF} = 7$, the largest schedulable transport block is TBS = 1736 bits. On the other hand, for "stand-alone and guard-band deployments" the transport block index ranges from $I_{TBS} = 0$ to $I_{TBS} = 13$, whereas the subframe index ranges from $I_{SF} = 0$ to $I_{SF} = 7$, the largest schedulable transport block is TBS = 2536 bits. In Rel-17, 16-QAM in both UL and DL was standardized with different design guidelines for UL and DL as described below.

Technical Background on LTE-M

LTE-M introduced a low-complexity User Equipment (UE) Category M1 (including half-duplex operation and lower power classes) that with respect to LTE UE Category 1 provides reduction in terms of peak rate, bandwidth, and number of antennas to lower the device's cost while targeting a reduced power consumption through extended discontinuous reception (eDRX) cycles, and two coverage enhancement (CE) modes. CE mode A is mandatorily supported to entitle moderately enhanced coverage, whereas CE mode B support is optional, targeting deep coverage. The physical layer enhancement for LTE-M in Rel-17 was aimed at increasing the peak data rates through the support of up to 14 HARQ processes and a larger maximum DL TBS for HD-FDD Cat-M1 UEs; hence, in the subsections below, we describe the baseline until Rel-16 in relation to the Rel-17 enhancements for LTE-M.

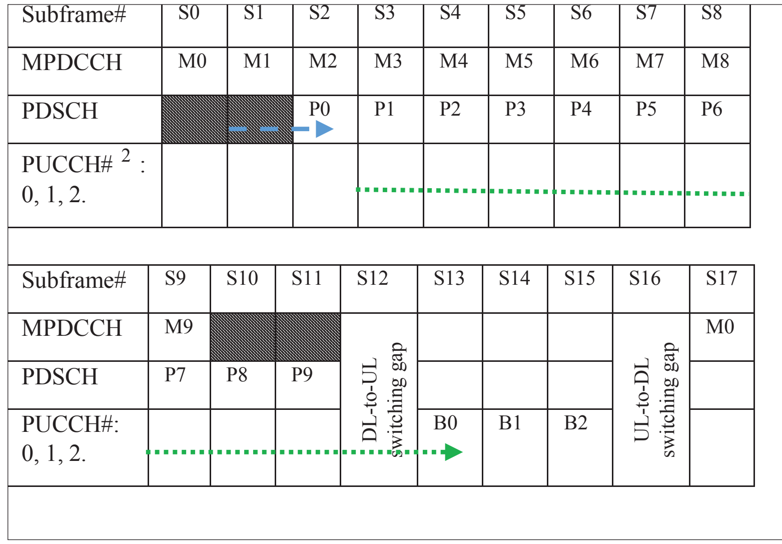

Figure 1. 10 HARQ processes and HARQ-ACK bundling combined.

LTE-M: 10 HARQ Processes and HARQ-ACK Bundling for HD-FDD Cat-M1 UEs: In Rel-14, two features that complement each other were introduced: The possibility of using up to 10 HARQ processes in DL along with the HARQ-ACK bundling feature, which basically allows the positive or negative acknowledgment (ACK/NACK) feedback of up to four different HARQ processes to be bundled into a single physical uplink control channel (PUCCH) transmission. Figure 1 illustrates the presence of 10 HARQ processes using the HARQ-ACK bundling feature.

In Fig. 1, M0 and P0 refer to the control channel (i.e., MTC physical downlink control channel [MPDCCH]) and corresponding user data (i.e., physical downlink shared channel [PDSCH]) associated with HARQ Process #0 which are transmitted in subframes S0 and S2, respectively. S12 is used by the UE as a guard-time to perform DL-to-UL switching; thereafter, PUCCH #0 represented as B0 in S13 bundles three ACKs/NACKs associated to multiple HARQ processes,[2] and S16 is used as another guard-time to perform UL-to-DL switching. Other HARQ processes in the diagram follow the same logic and corresponding terminology. Moreover, the blue dashed and green dotted arrows refer to the PDSCH scheduling delay and HARQ-ACK delay respectively.

PDSCH Scheduling Delay: It refers to the number of subframes that need to pass before transmitting the user data once the control information has been transmitted. The PDSCH scheduling delay has been defined as 2 BL/CE[3] DL subframes. For example, the blue dotted arrow in Fig. 1 illustrates the PDSCH scheduling delay

[2] ACK/NACK as bundled on PUCCH#: 0{0, 1, 2}, 1{3, 4, 5}, and 2{6, 7, 8, 9}. The numbers inside the curly brackets refer to the HARQ processes into which ACK/NACK is being bundled.

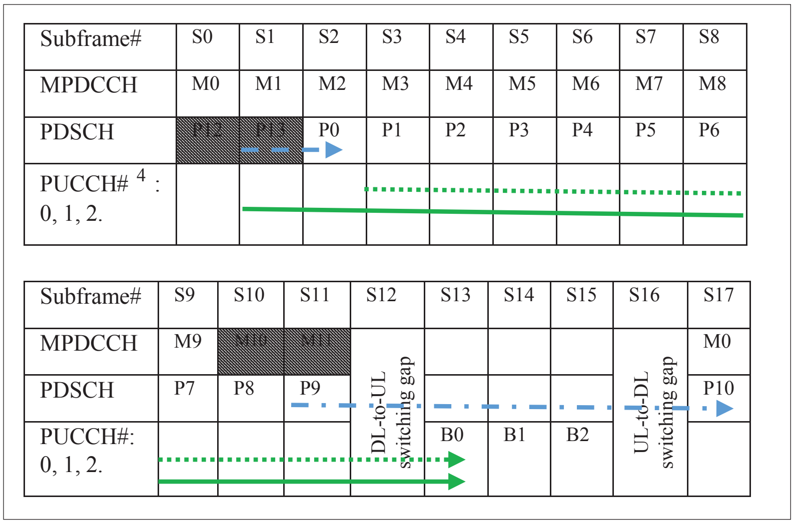

FIGURE 2. 14 HARQ processes and HARQ-ACK bundling combined.

for the first HARQ process; that is, once M0 was transmitted in S0, subframe S1 is the 1st out of 2 BL/CE subframes, whereas S2 is the 2nd out of 2 BL/CE DL subframes where P0 is transmitted.

HARQ-ACK Delay: It refers to the number of subframes that need to pass before transmitting the ACK/NACK once the corresponding user data has been transmitted. The HARQ-ACK delay has been defined in terms of absolute subframes, being any of the values in the HARQ-ACK delay set = {4, 5, 6, 7, 8, 9, 10, 11}, a selectable delay. For example, the green dotted arrow in Fig. 1 illustrates the HARQ-ACK delay for the first HARQ process: once P0 was transmitted in S2, subframe S3 is the 1st out of 11 absolute subframes, and the counting continues until S13, which is the 11th out of 11 absolute subframes where the ACK/NACK for P0 is transmitted over PUCCH#0.

Thus, assuming a TBS = 1000 bits and 10 HARQ processes along with HARQ-ACK bundling, a peak data rate of 588 kb/s can be achieved. In Rel-17, the unused subframes in the legacy framework (i.e., shadowed subframes to the left of P0 and to the right of M9 in Fig. 1) were exploited to introduce the use of up to 14 HARQ processes as elaborated below.

LTE-M: Max DL TBS for HD-FDD Cat-M1 UEs: The maximum DL TBS for HD-FDD Cat-M1 UEs is 1000 bits, since larger transport block sizes (available in the TBS/MCS tables reused from LTE) are not selectable for HD-FDD Cat-M1 UEs. As part of the Rel-17 preparations, a set of use cases were discussed to justify the support of a larger DL TBS. The discussed use cases included "watches and wearables," "high data rate music," "small-format video," "firmware download," and "point-of-sale transactions." As a conclusion, watches/wearables and small-format video, which require a peak data rate of around 1 Mb/s in the physical layer, were identified as relevant use cases to be delivered by an HD-FDD Cat-M1 UEs [6]. Thus, in Rel-17 a maximum DL TBS of 1736 bits was made available as an enhancement for HD-FDD Cat-M1 UEs in CE mode A, which required a soft-buffer size recalculation to provide higher throughput (around 73.6 percent increase) and other benefits described below.

[3] BL/CE: Bandwidth-reduced low-complexity or coverage enhanced.

[4] ACK/NACK as bundled on PUCCH#: 0{12, 13, 0, 1}, 1{2, 3, 4, 5}, and 2{6, 7, 8, 9}. The numbers inside the curly brackets refer to the HARQ processes which ACK/NACK is being bundled.

3GPP Rel-17 Physical Layer Enhancements

NB-IoT: 16-QAM in UL and DL

The introduction of 16-QAM in DL overall considered the following design guidelines [4]:

- Increase the maximum TBS (e.g., 2× compared to Rel-16) and soft-buffer size.
- Perform required modifications on the DL power control equation for NPDSCH.
- Extend the channel quality reporting to support 16-QAM.

The initial step toward introducing 16-QAM in DL was to build the TBS/MCS table accounting for achievable code rates (including standalone/guard-band and in-band deployments), avoidance of link adaptation issues (which refers to avoiding unstable performance between adjacent operation points in the TBS/MCS table), breaking point between modulation schemes, downlink control information (DCI) impact (e.g., how to indicate the usage of 16-QAM), and so on. The TBS/MCS table for 16-QAM in DL can be found in [5]. For standalone/guard-band deployments, the TBS/MCS table ranges from I_{TBS} = 14 to I_{TBS} = 21, and from I_{SF} = 0 to I_{SF} = 7; the largest schedulable TBS = 4968 bits. For in-band deployment, the TBS/MCS table ranges from I_{TBS} = 11 to I_{TBS} = 17, and from I_{SF} = 0 to I_{SF} = 7; the largest schedulable TBS = 3624 bits. The largest TBS usable with 16-QAM in DL (i.e., TBS = 4968 bits for standalone/guard-band deployments) previously had been used in LTE and was the closest value to twice the maximum TBS available in Rel-16. Later, we present the 10 percent BLER performance for all the TBS entries usable for 16-QAM in DL for standalone/guard-band deployments. Moreover, 16-QAM NPDSCH transmissions do not support the use of repetitions. The DL power allocation accounts for the differences in the deployment modes and includes signaling of data-to-pilot power ratios as described in [5]. The DL channel quality reporting for 16-QAM is based on NPDSCH transport blocks achieving an error probability not exceeding 10 percent BLER; the channel quality information (CQI) mapping table including the 16-QAM reports can be found in [7]. The soft-buffer size for supporting 16-QAM consists of 12,800 bits (i.e., the size was doubled with respect to the size used for quadrature phase shift keying [QPSK]) [8].

The introduction of 16-QAM in UL considered a single design guideline [4]: The maximum TBS available in legacy (i.e., Rel-16) is not increased. The same technical considerations as in DL were used for designing the TBS/MCS table, except that in UL there is no need to consider different I_{TBS} ranges as a function of deployment modes. The TBS/MCS table for 16-QAM in UL can be found in [5], and ranges from I_{TBS} = 14 to I_{TBS} = 21, and from I_{RU} = 0 to I_{RU} = 6; the largest schedulable TBS = 2536 bits.

For 16-QAM in UL, aiming at doubling the throughput with respect to QPSK, the TBS/MCS table was designed to allow transmitting the TBS = 2536 bits with half the number of RUs as compared to QPSK when scheduling ("I_{TBS}=21," "I_{RU}=4"). Moreover, 16-QAM NPUSCH format 1 transmissions do not support the use of repetitions, so a new term (i.e., Δ_{TF}) in the NPUS-

CH power control equation was introduced to account for the higher modulation scheme [5], and only multi-tones (3, 6, and 12 subcarriers) are schedulable for 16-QAM in UL.

LTE-M: 14 HARQ Processes for HD-FDD Cat-M1 UEs

In Rel-17, the number of HARQ processes were increased from 10 to 14 for an HD-FDD Cat-M1 UE. The framework utilized to introduce 10 HARQ processes left two subframes per scheduling cycle unused (i.e., every 10 HARQ processes, there are two subframes with the potential to be used for transmitting user data in DL). The introduction of 14 HARQ processes aimed to exploit those unused subframes (i.e., shadowed subframes to the left of P0 and to the right of M9 in Fig. 1), increasing the peak data rate from 588 kb/s to 706 kb/s.

From Fig. 2, it can be observed that the two subframes to the right of M9 in S9 (i.e., shadowed subframes) were utilized to incorporate two additional HARQ processes through M10 and M11 with their associated P10 and P11 being transmitted along with their corresponding ACK/NACK in the subsequent scheduling cycle. Such a scheduling cycle starts from S17 upon transmitting M0, M1, …, M9 along with another two HARQ processes corresponding to M12 and M13. In other words, in Rel-17 it is possible to use up to 14 HARQ processes through scheduling 12 HARQ processes per scheduling cycle (i.e., M0 to M9 appending in an alternating manner, either "M10 and M11" or "M12 and M13") exploiting the unused resources in the legacy framework. This is possible because the HARQ-ACK bundling allows up to 4 ACKs/NACKs to be bundled per PUCCH, which means that 3 PUCCHs are sufficient to support 12 HARQ processes per scheduling cycle. Moreover, the support of 14 HARQ processes has side implications on both the PDSCH scheduling delay and HARQ-ACK delay. In Fig. 2, the blue dash-dotted arrow illustrates that a PDSCH scheduling delay longer than 2 BL/CE DL subframes is required (i.e., ideally a delay of 7 BL/CE DL subframes is required). Similarly, for the HARQ-ACK delay, the green solid arrow in Fig. 2 shows that the maximum available delay in the HARQ-ACK delay set that equals 11 subframes is no longer sufficient (i.e., ideally 13 subframes are required). Figure 2 assumes a scenario where all the subframes are valid subframes; hence, it depicts the ideally required delays. A subframe is determined to be a valid subframe or an invalid subframe using DL and UL bitmaps configured via network's higher layer signaling (i.e., radio resource control [RRC] signaling). In an invalid DL subframe (i.e., non-BL/CE DL subframe[5]), an HD-FDD Cat-M1 UE can only transmit in UL or perform either a DL-to-UL switching or a UL-to-DL switching, whereas in an invalid UL subframe (i.e., non-BL/CE UL subframe) an HD-FDD Cat-M1 UE can optionally transmit PUCCH without repetition. Otherwise, it can only transmit in DL or perform either a DL-to-UL switching or a UL-to-DL switching. Real network deployments usually configure invalid subframes, and because of that in Rel-17 both the PDSCH scheduling delay and the HARQ-ACK delay were enhanced.

PDSCH Scheduling Delay: A PDSCH scheduling delay expressed in terms of several subframe types was standardized as described in Table 5.3.3.1.12-3 of 3GPP Technical Specification (TS) 36.212 [9].

HARQ-ACK Delay: For the HARQ-ACK delay, two solutions were standardized either of them to be used based on the configuration provided via RRC signaling:

- HARQ-ACK delay solution 1: It is determined through an expression consisting of different subframe types (using a similar principle as for the PDSCH scheduling delay) as described in Table 5.3.3.1.12-2 of [9]. This future-proof solution envisions a large presence of invalid subframes, which will allow, for example, more efficient coexistence with 5G NR.
- HARQ-ACK delay solution 2: The HARQ-ACK delay is expressed in terms of absolute subframes, which includes the minimum delay required (13 subframes) to support 14 HARQ processes and other delays (see the full HARQ-ACK delay set in Table 5.3.3.1.12-1 of [9]). This solution is intended to be used for scenarios when there is low (or no) presence of invalid subframes.

The PDSCH scheduling delay solution can be used along with either of the HARQ-ACK delay solutions configured using RRC signaling. Moreover, the PDSCH scheduling delay and HARQ-ACK delay are jointly encoded as described in [9] and are provided to the UE using physical layer signaling on MPDCCH. Figure 3 depicts a scenario with 14 HARQ processes in the presence of invalid subframes according to the DL bitmap "1 1 1 1 1 1 1 0 0 1," where for illustration purposes "0" refers to a non-BL/CE DL subframe.

To illustrate how the 14 HARQ processes feature operates in the presence of non-BL/CE DL subframes, let us focus on one of the newly added HARQ processes, for example, the 11th HARQ process scheduled with M10 in subframe S12. The PDSCH scheduling delay expressed in terms of different subframe types is used to avoid skipping subframes unnecessarily (which otherwise would result in a loss of throughput); hence, for the 11th HARQ process under analysis, the PDSCH scheduling delay option 1 in Table 5.3.3.1.12-3 of [9] applies, where the first term in the expression (i.e., 1 BL/CE DL subframe) is counted in S13, the second term (i.e., 1 subframe) is counted in S14, the third term (i.e., 3 BL/CE UL subframes) is counted through S15, S16, and S17, respectively, the fourth term (i.e., 1 subframe) is counted in S18, and finally, the fifth term (i.e., 1 BL/CE DL subframe) takes us to transmit P10 in S19 to provide a peak data rate of 631.5 kb/s. If instead the PDSCH scheduling delay had been defined simply as 7 BL/CE DL subframes, a lower peak data rate of 571.5 kb/s would have been achieved since P10 would have been transmitted in S21. The HARQ-ACK delay as in Table 5.3.3.1.12-2 of [9] required in this case $y = 11$ and $z = 1$. Later, we present a table with more peak data rates estimates as a function of the presence of invalid subframes.

LTE-M: Max DL TBS of 1736 Bits for HD-FDD Cat-M1 UEs

In Rel-17, the maximum DL TBS for HD-FDD Cat-M1 UEs in CE mode A was increased from 1000 bits to 1736 bits. Equation 1 was used to calculate the required number of soft channel bits.

The initial step toward introducing 16-QAM in DL was to build the TBS/MCS table accounting for achievable code rates (including standalone/guard-band and in-band deployments), avoidance of link adaptation issues (which refers to avoiding an unstable performance between adjacent operation points in the TBS/MCS table), breaking point between modulation schemes, downlink control information (DCI) impact (e.g., how to indicate the usage of 16-QAM), and so on.

[5] The terms "non-BL/CE DL subframes" and "non-BL/CE UL subframes" were recently updated in the 3GPP specifications as "DL subframes that are not BL/CE DL subframes" and "UL subframes that are not BL/CE UL subframes".

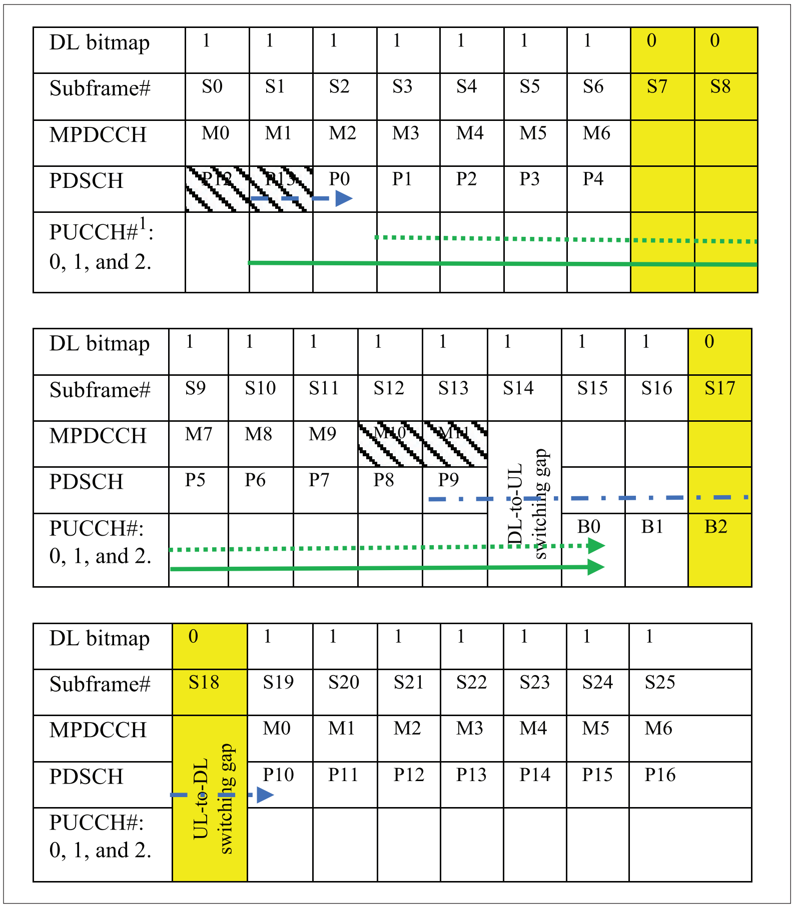

Figure 3. 14 HARQ processes and HARQ-ACK bundling combined in presence of invalid downlink subframes.

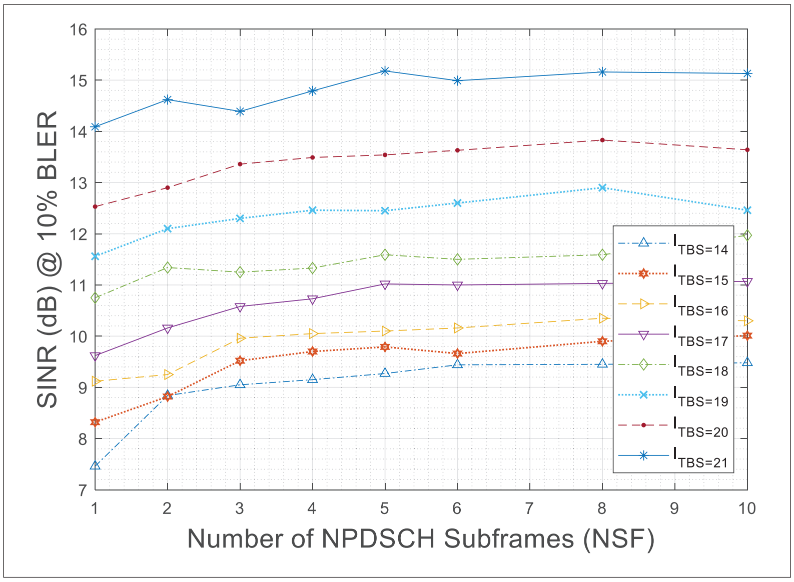

Figure 4. 10 % BLER TBS/MCS table for standalone/guard-band deployments.

$$\textit{Soft Buffer Size} = (N)(32)(3)(\lceil (X + 24 + 4)/32 \rceil) \quad (1)$$

In Eq. 1, *N* is the number of HARQ processes, 32 ensures that the soft buffer size is a multiple of 32 bits, 3 relates to the coding rate of the turbo encoder, 24 relates to the cyclic redundancy check (CRC) size, 4 relates to the tail bits of the turbo encoder [9], and X is the TBS (X ≤ 6120). Moreover, ⌈.⌉ is the ceiling function, which rounds the value toward positive infinity. In Rel-17, X is the new maximum DL TBS of 1736 bits, whereas to find a suitable value for *N*, a set of link-level simulations were performed comparing throughput vs. signal-to-noise ratio (SNR) assuming different soft buffer sizes. As shown below, *N* = 8 was identified as a suitable value, which allowed using Eq. 1 to determine a soft buffer size of 43,008 soft channel bits. Moreover, it was concluded that the new maximum DL TBS of 1736 bits is compatible with all other LTE-M features applicable for HD-FDD Cat-M1 UEs in CE mode A, and that the feature will be enabled using unicast RRC signaling.

[6] ACK/NACK as bundled on PUCCH#: 0{12, 13, 0, 1}, 1{2, 3, 4, 5}, and 2{6, 7, 8, 9}. The numbers inside the curly brackets refer to the HARQ processes which ACK/NACK is being bundled.

Evaluation Results

Below, we present evaluation results for the physical layer enhancements introduced for LTE-M and NB-IoT in Rel-17.

NB-IoT: 16-QAM in UL and DL

Figure 1 shows the 10 percent BLER performance for all transport blocks in the TBS/MCS table standardized for 16-QAM in DL applicable for standalone/guard-band deployments [9], where BLER is defined as the ratio between the incorrectly received transport blocks and all the transport blocks sent. The simulation assumptions can be found in [10].

In Fig. 4, each operation point resembles the performance of the transport block entries in the TBS/MCS table for standalone/guard-band deployments, which spans from I_{TBS} = 14 to I_{TBS} = 21 and from I_{SF} = 0 to I_{SF} = 7 [5]. Figure 4 already includes in the x-axis the mapping of subframe index (i.e., I_{SF}) to number of NPDSCH subframes [5]. The performance in Fig. 4 shows that the curves from the smallest schedulable TBSs (I_{TBS} = 14) to the largest schedulable TBSs (I_{TBS} = 21) appear in ascending order without incurring any performance crossing, which reflects no link adaptation issues.

LTE-M: 14 HARQ Processes for HD-FDD Cat-M1 UEs

Previously, the 14 HARQ processes feature for HD-FDD Cat-M1 UEs was described, which was designed to avoid unnecessary loss of throughput due to the presence of invalid subframes. Table 1 summarizes the achievable peak data rates when there are 14 HARQ processes in use in the presence of different amounts of invalid subframes.

In Table 1, DL transmissions are assumed to start at the subframe where the leftmost bit in the bitmap applies. The presence of invalid subframes assumed in Table 1 results in an inevitable loss of throughput, which is kept to what is strictly unavoidable through the PDSCH scheduling delay and HARQ-ACK delay solutions standardized in Rel-17 for the 14 HARQ processes feature [9]. The enhanced PDSCH scheduling delay and HARQ-ACK delay allow transmitting at the most immediately available valid subframe to avoid any unnecessary loss of throughput; otherwise, the observable loss of throughput would be larger than what is inherently unavoidable. Moreover, Table I includes estimates for a TBS equal to 1736 bits since this is the new maximum DL TBS that became available for HD-FDD Cat-M1 UEs in Rel-17.

	Scenarios without presence of invalid subframes	Number of invalid subframes (denoted by 0 in the bitmaps) cyclically present every 10 ms, causing a transmission postponement		
		DL bitmap: "1 1 1 1 1 1 1 0 0 1."	DL bitmap: "1 1 1 1 1 1 0 0 0 0."	DL bitmap: "1 1 1 1 1 1 0 0 0 0." UL bitmap: "1 1 1 1 1 1 1 1 0 1."
TBS = 1000 bits	~ 706 kb/s	~ 632 kb/s	~ 571 kb/s	~ 545 kb/s
TBS = 1736 bits	~ 1.225 Mb/s	~ 1.096 Mb/s	~ 992 kb/s	~ 947 kb/s

Table 1. Achievable peak data rates over one scheduling cycle: 14 HARQ processes with and without presence of invalid subframes.

LTE-M: Max DL TBS of 1736 Bits for HD-FDD Cat-M1 UEs

The support of a new maximum DL TBS of 1736 bits for HD-FDD Cat-M1 UEs in CE mode A required an increase in the soft-buffer size, which calculation depends on the number of assumed HARQ processes (N in Eq. 1). In Fig. 5, we present link-level simulations comparing throughput vs. SNR assuming different soft buffer sizes.

From Fig. 5, the achievable throughput assuming 8 HARQ processes (43,008 soft channel bits) is comparable to the performance when the assumption is 10 HARQ processes (53,760 soft channel bits), whereas using a number of HARQ processes < 8 HARQ processes (e.g., N = 6 to obtain 32,256 soft channel bits or N = 4 to obtain 21,504 soft channel bits) results in throughput degradation. Thus, using N = 8 (i.e., 8 HARQ processes) is a suitable choice for TBS = 1736 bits.

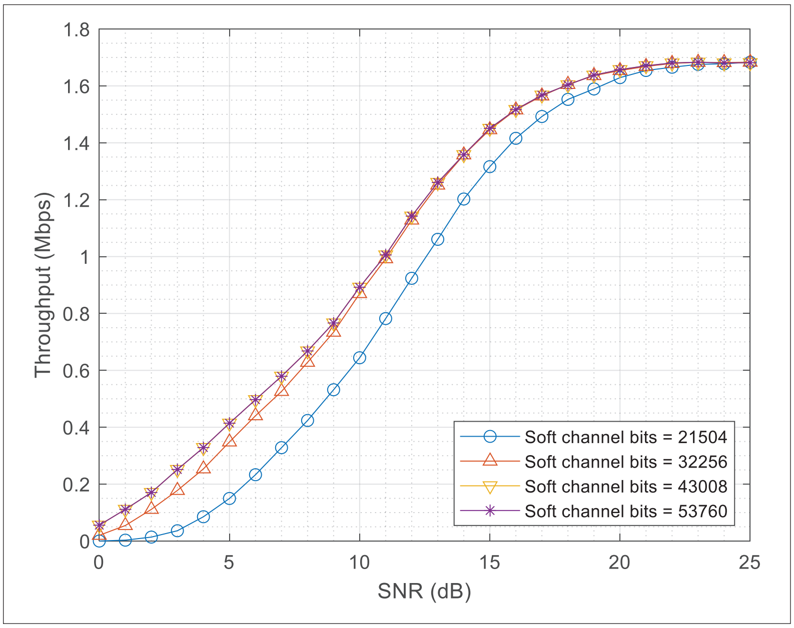

Figure 5. Throughput for different soft-buffer sizes from 4, 6, 8, and 10 HARQ processes, 16-QAM, and a 4-physical resource block (PRB) allocation.

Conclusions and Future Work

In this article, we provide an overview of the Rel-17 physical layer enhancements standardized for NB-IoT and LTE-M:

- 16-QAM in UL and DL for NB-IoT: In DL, a TBS of 4968 bits was introduced to provide ~ 2× the maximum achievable throughput with respect to QPSK. The TBS/MCS tables for standalone/guard-band and in-band deployments consist of 64 TBS entries and 56 TBS entries, respectively. Moreover, 16-QAM in DL used at most one repetition and included support for channel quality reporting and DL power allocation. On the other hand, the design guideline for 16-QAM in UL did not allow increasing the largest TBS (2536 bits) available for QPSK, so such a TBS was designed to be transmitted using half of the time-domain resources. Moreover, the TBS/MCS table for UL has 49 TBS entries that are applicable for all deployment modes, at most one repetition is used, multi-tone allocation is supported (3, 6, and 12 subcarriers), and one power boosting term (i.e., Δ_{TF}) was added into the UE's transmit power control equation.
- 14 HARQ processes in DL for HD-FDD Cat-M1 UEs exploits the unused transmission opportunities in the 10 HARQ processes/HARQ bundling framework to add four additional HARQ processes, which increases the peak data rate from 588 kb/s to 706 kb/s. Moreover, the PDSCH scheduling delay and HARQ-ACK delay were designed to deal with the presence of invalid subframes to avoid unnecesary loss of throughput derived from skipping unnecessarily valid subframes.
- A new maximum DL TBS of 1736 bits was standardized for HD-FDD Cat-M1 UEs in CE Mode A, which provides ~ 73.6 percent throughput increase. The new TBS is applicable for all HD-FDD Cat-M1 UEs features in CE mode A.
- In terms of future work:
  –For 16-QAM for NB-IoT, enabling an earlier usage of 16-QAM transmissions can be considered, for example, during random access procedure.
  –The 14 HARQ processes framework for LTE-M could be enhanced to reduce the unnecessary loss of throughput derived from the presence of measurement gaps.

Finally, LTE-M and NB-IoT are nowadays envisioned to be focused on adding support for non-terrestrial networks.

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